

Geometric analysis of nonlinear impulsive control systems: Decompositions, accessibility, observability

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ARTICLE INFO

Keywords:

Differential geometric approach
Impulsive control systems
Decompositions
Accessibility
Observability

ABSTRACT

This paper investigates some fundamental properties of nonlinear impulsive control systems (ICS), including structural decompositions, accessibility and observability. By defining suitable invariant distributions using differential geometry methods, we propose the controllability decomposition and the observability decomposition for nonlinear ICS. With the help of such structural decompositions, the geometrical characterizations of the reachable set and the indistinguishable set of nonlinear ICS are presented. The corresponding criteria for (strong) accessibility and observability of nonlinear ICS are obtained. For a special class of linear ICS, we show that the strong accessibility is equivalent to the controllability. Finally, examples illustrate the effectiveness of the main theoretical results.

1. Introduction

The analysis of the fundamental properties of nonlinear control systems has been a hot topic over the past few decades [1–3]. Controllability is a fundamental structural property for linear control systems, which is about the possibility of steering the systems from the initial state to the final state. Given the complexity of nonlinear control systems, the property of such systems that is similar to controllability in linear control systems is accessibility, which is the property that one should be able to reach a set of full dimension from any initial state [3]. Observability is another fundamental structural property for nonlinear control systems. Roughly speaking, it is related to the concept of indistinguishability of states with respect to the inputs [4]. In order to study systems that are not completely controllable or observable, Kalman canonical decomposition was proposed for linear control systems [5] and was generalized to nonlinear control systems [2].

Impulsive control systems (ICS) are important types of hybrid control systems, which have attracted much attention due to their wide range of applications in various fields, such as time-delay systems [6], stochastic systems [7], fully-actuated systems [8], hybrid descriptor systems [9]. Furthermore, impulsive control theories were used in many real world systems. For example, the problem of stabilization for multi-agent systems over finite fields via impulsive control was investigated in [10]. Ref. [11] studied the controllability property of multi-agent networks with delayed input and impulse. Ref. [12]

presented the stability analysis and comparative study on different eco-epidemiological models with impulsive effects. The authors in [13] studied the robust exponential input-to-state stability for impulsive systems, which was used to test the input-to-state stability for a controlled micro-grid.

Some properties such as controllability, reachability and observability of linear ICS, have been extensively studied during the last decades. For example, in [14], some criteria for controllability and observability were established for a class of linear ICS. In [15], concepts of reachability and observability for linear ICS were addressed and geometric characterizations of the reachable and unobservable sets were presented. Ref. [16] reported the controllability and observability of piecewise linear time-varying ICS. In [17], the controllability of linear stochastic ICS in Hilbert spaces was studied.

In this paper, we will investigate three fundamental properties (decompositions, accessibility, observability) for nonlinear ICS. It is noted that, the general solution can be given explicitly with the help of the state transition matrix for linear ICS, which is the main means of studying fundamental properties of such systems, such as [14–17]. For semilinear ICS, with the help of this approach, there are some recent results on the fundamental properties [18,19]. Since this approach is difficult to realize in nonlinear ICS discussed in this paper, the transition from studying fundamental properties of linear ICS to nonlinear ICS poses a significant challenge. Algebraic methods are currently more commonly applied for the study of fundamental properties of

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nonlinear ICS. By algebraic methods, the authors in [20] generalized the controllability property of linear ICS in [21] and studied the accessibility of nonlinear ICS. Then, they investigated observability of nonlinear ICS for the first time in [22]. Geometric methods, as distinct from algebraic methods, are also important in the study of nonlinear control systems [1]. Ref. [15] firstly used geometric methods to study the reachability and observability for linear ICS. Ref. [23] used the differential geometric analysis and Lie group theory to investigated the accessibility and observability for discrete nonlinear ICS. The authors in [24] introduced the concept of distributions for nonlinear ICS, which is the basic tool in the study of nonlinear control systems by geometric methods [1].

Motivated by [15,23,24], this paper investigates the fundamental properties of nonlinear ICS using differential geometry methods. Note that the nonlinear ICS which used the impulsive input studied in [20,22] is different from the nonlinear ICS with the continuous input will be studied in this paper. From the perspective of the impulsive effects, the former studies the impulsive control problem, and the latter studies the impulsive perturbation problem that has received considerable critical attention such as [25–28]. Furthermore, the definitions of accessibility and observability in this paper are inspired by [1,2], which are different from those in [20,22] which adopt an algebraic perspective. Notably, algebraic criteria for testing the accessibility and observability were obtained in [20,22]. In contrast to these algebraic methods, this paper utilizes geometric approaches to characterize the reachable and indistinguishable sets. These characterizations have close connections to realization theory, pole assignment, structural decomposition, observer design, etc; see [29] and the reference therein. Besides, different from [24], this paper considers two equivalent forms of the nonlinear ICS. It provides a more detailed discussion on the relationship between the local coordinate transformation function and the impulsive function. To the best of our knowledge, there have been no result on the decompositions, accessibility and observability of nonlinear ICS from a geometric perspective. It is meaningful and challenging to develop a theoretical framework for the geometric analysis of nonlinear ICS.

The main contributions of the paper are threefold: (i) Two kinds of structural decompositions for nonlinear ICS are proposed, providing a basis for (strong) accessibility and observability analysis. (ii) The geometric characterizations of reachable set and the indistinguishable set are presented. The (strong) accessibility and the observability of nonlinear ICS are investigated and the corresponding rank conditions are obtained in this paper. (iii) Some theoretical results are used in linear ICS, and we show that for a class of linear ICS, the strong accessibility is equivalent to the controllability.

2. Preliminary

Notations. $\mathbb{Z}_+$ is the set of positive integers, $\mathbb{R}^n$ is the n -dimensional real space. $[f, g]$ is the Lie bracket of the two vector fields f and g . $L_f g$ is the Lie derivative of g along f . $\text{ad}_f^0 g = g$ and $\text{ad}_f^k g = [f, \text{ad}_f^{k-1} g]$, $k \in \mathbb{Z}_+$. Let $\mathcal{R}$ be a set, and $\text{int}(\mathcal{R})$ denote the interior points of $\mathcal{R}$. Denote $\text{col}(\cdot)$ the operation of forming a column vector by stacking its elements vertically. Let $\Phi_f^t(x)$ denote the flow of the vector field f .

In this paper, we consider the nonlinear ICS with m inputs $u_1, \dots, u_m \in \mathbb{R}$ and p outputs $y_1, \dots, y_p \in \mathbb{R}$ in a form like:

$$\begin{cases} \dot{x}(t) = f(x(t)) + \sum_{i=1}^m g_i(x(t))u_i(t), & t \neq \tau_k, \\ x(\tau_k^+) = \varphi(x(\tau_k)), & t = \tau_k, \\ y_j(t) = h_j(x(t)), & 1 \leq j \leq p. \end{cases} \quad (1)$$

where $t \geq 0$ denotes time, $x \in U \subseteq \mathbb{R}^n$ denotes the state variable defined on an open set U . $f, g_1, \dots, g_m, \varphi$ are smooth vector fields defined on U , and $h_1, \dots, h_p$ are smooth real-valued functions defined on U . In this paper, we consider admissible control input $u_i, i = 1, \dots, m$ are piecewise constant functions. $\mathcal{T} = \{\tau_k, k \in \mathbb{Z}_+\}$ is the set of impulse

times satisfying $0 < \tau_1 < \dots < \tau_k < \dots, \lim_{k \rightarrow \infty} \tau_k = \infty$. Here, we assume that the state variables of system (1) are left continuous at each τ_k , i.e., $x(\tau_k) = x(\tau_k^-)$, and limited on the right. Note that the system (1) can be written alternatively as [22,30]:

$$\begin{cases} \dot{x}(t) = f(x(t)) + \sum_{i=1}^m g_i(x(t))u_i(t) + \sum_{k=1}^{\infty} \varphi^+(x(t))\delta(t - \tau_k), \\ y_j(t) = h_j(x(t)), 1 \leq j \leq p. \end{cases} \quad (2)$$

where $\varphi^+(x) = \varphi(x) - x$ and δ is the impulse applied at $t = \tau_k, k \in \mathbb{Z}_+$. We can obtain that $\varphi^+(x(\tau_k)) = \varphi(x(\tau_k)) - x(\tau_k) = x(\tau_k^+) - x(\tau_k)$.

It is noted that, for purely continuous system defined on $\mathbb{R}^n$, the mappings $f, g_1, \dots, g_m$ are smooth mappings assigning to each point x of U a vector of $\mathbb{R}^n$, namely $f(x), g_1(x), \dots, g_m(x)$. For this reason, they are frequently referred to as vector fields defined on U [1]. In this article, the concept of vector fields is extended to impulsive systems. Since φ and φ^+ are smooth mappings assigning to each point x of U a vector of $\mathbb{R}^n$, they are also referred to as vector fields, which can also be seen in [31].

We next give the notion of distribution, which plays a fundamental role in geometrical analyzing of nonlinear control systems [1]. Suppose now that d smooth vector fields $f_1, \dots, f_d$ defined on the same open subset $U \subseteq \mathbb{R}^n$ are given. At any fixed point x in U , the vectors $f_1(x), \dots, f_d(x)$ span a vector space. Let the symbol $\Delta(x)$ denote this vector space, namely $\Delta(x) = \text{span}\{f_1(x), \dots, f_d(x)\}$. Note that, in doing this, we have essentially assigned a vector space to each point x of the set U . The assignment described above is called a smooth distribution. According to the characterization just given, a distribution is identified by a set of vector fields $f_1, \dots, f_d$, and we will use the notation $\Delta = \text{span}\{f_1, \dots, f_d\}$ to denote the assignment as a whole, and $\Delta(x)$ to denote the “value” of Δ at a point x .

In the following, some definitions and a lemma of distribution are reviewed.

Definition 1 ([1]). Suppose that a smooth distribution Δ defined on a subset $U \subseteq \mathbb{R}^n$ is given. Then, Δ is said to be

- (i) *nonsingular* if there exists an integer $d \geq 0$ such that $\dim \Delta(x) = d$ for any $x \in U$. A point $x_0 \in U$ is said to be a *regular* point of Δ , if there exists a neighborhood U_0 of x_0 such that Δ is nonsingular on U_0 .
- (ii) *involutive* if the vector field $[f_1, f_2]$ belongs to Δ for any pair of vector fields $f_1, f_2 \in \Delta$.
- (iii) *invariant* under the vector field f , if the Lie bracket $[f, \Delta]$ is contained in Δ , i.e., if $[f, \Delta](x) \subseteq \Delta(x)$ for all $x \in U$. The notation $[f, \Delta]$ denotes the distribution consisting of all the vector fields of the form $[f, f_1]$, where f is the given vector field and $f_1 \in \Delta$.

The dual notion of the co-distribution is also be considered that is defined in the following way. Suppose now that d smooth covector fields $w_1, \dots, w_d$ defined on the open subset $U \subseteq \mathbb{R}^n$ are given, one can associate the assignment, to each point $x \in U$, a subspace of $(\mathbb{R}^n)^*$, the one spanned by the covectors $w_1, \dots, w_d$. The object characterized in this way is called a smooth co-distribution. Coherently with the notations introduced for distributions, we use the notation $\Omega = \text{span}\{w_1, \dots, w_d\}$ to denote the assignment as a whole, and $\Omega(x)$ to denote the “value” of Ω at a point x . We sometimes construct the distribution starting from the co-distribution, so it necessary to consider the annihilator of Ω , that is the distribution $\Omega^\perp$ defined by $\Omega^\perp(x) = \{v \in \mathbb{R}^n : \langle w^*, v \rangle = 0, \forall w^* \in \Omega(x)\}$. Conversely, given a distribution Δ , one can construct a co-distribution, noted $\Delta^\perp$, setting at each x in U : $\Delta^\perp(x) = \{w \in (\mathbb{R}^n)^* : \langle w^*, v \rangle = 0, \forall v \in \Delta(x)\}$.

Definition 2 ([1]). A nonsingular d -dimensional distribution Δ , defined on an open set U , is said to be *completely integrable* if for each point x_0 of U there exist a neighborhood U_0 of x_0 , and $n - d$ real-valued smooth functions $\lambda_1, \dots, \lambda_{n-d}$ defined on U_0 , such that $\Delta^\perp(x) = \text{span}\{d\lambda_1, \dots, d\lambda_{n-d}\}$ on U_0 .

Lemma 1 ([1]). A nonsingular distribution is completely integrable if and only if it is involutive.

A key benefit of complete integrability is that, it can define (locally around a point x_0) a coordinates transformation entailing a particularly simple representation for the vector fields of Δ [1]. Besides, in this paper, we always consider the coordinates transformation $z = \Psi(x)$ with the following assumption:

Assumption 1 ([1]). $\Psi(x)$ is invertible and $\Psi(x), \Psi^{-1}(z)$ are both smooth mappings.

By Assumption 1, the description of the system in the new coordinates will be still a smooth one.

3. Main results

3.1. Local decompositions

In many cases for nonlinear systems, one rather looks at transformations with local case [1], i.e. to restrict the attention only to a neighborhood U_0 of a given point x_0 . In this paper, due to the impulsive effects, the adoption of a local point of view needs some care. In facts, all trajectories starting from x_0 at $t = 0$ could leave U_0 at impulse times, making the local point of view inapplicable or deprived of practical interest [24]. Similar to [24], we make the following assumption.

Assumption 2. For the nonlinear ICS (1), we assume that in each impulse times, the impulses do not drive the state out of the considered neighborhood U_0 , i.e. by assuming that $\varphi(U_0) \subseteq U_0$.

Next we make the following assumption.

Assumption 3. Given a coordinates transformation $z = \Psi(x)$ for the nonlinear ICS (1) and (2). We assume that Ψ and $\varphi, \varphi^\dagger$ satisfies

$$\Psi \circ \varphi(\Psi^{-1}(z(\tau_k))) = z(\tau_k) + \frac{\partial \Psi}{\partial x} \varphi^\dagger(\Psi^{-1}(z(\tau_k))). \quad (3)$$

Assumption 3 considers that, the nonlinear ICS (1) and (2) are equivalent, they should also be equivalent after the coordinates transformation $z = \Psi(x)$. Specifically, we consider the left and right equations of (3). Denote $\bar{\varphi}$ and $\bar{\varphi}^\dagger$ the representation of φ and $\varphi^\dagger$ in the new coordinates, respectively. By $z = \Psi(x)$ and (1), we have

$$z(\tau_k^+) = \Psi(x(\tau_k^+)) = \Psi \circ \varphi(x(\tau_k)) = \Psi \circ \varphi(\Psi^{-1}(z(\tau_k))). \quad (4)$$

On the other hand, by $z = \Psi(x)$ and (2), we have

$$\begin{aligned} \dot{z}(t) &= \frac{\partial \Psi}{\partial x} \dot{x}(t) \\ &= \frac{\partial \Psi}{\partial x} (f(x(t)) + \sum_{i=1}^m g_i(x(t))u_i(t) + \varphi^\dagger(x(t)) \sum_{k=1}^{\infty} \delta(t - \tau_k)). \end{aligned} \quad (5)$$

Hence, we obtain

$$\bar{\varphi}^\dagger(z) = \frac{\partial \Psi}{\partial x} \varphi^\dagger(\Psi^{-1}(z)), \quad (6)$$

and

$$z(\tau_k^+) = \bar{\varphi}^\dagger(z(\tau_k)) + z(\tau_k) = \frac{\partial \Psi}{\partial x} \varphi^\dagger(\Psi^{-1}(z)) + z(\tau_k) \quad (7)$$

Hence, if Assumption 3 is satisfied, combining (5) and (7), the nonlinear ICS (1) and (2) will be equivalent after the coordinates transformation $z = \Psi(x)$.

From Assumption 3, the selection of coordinate transformations for nonlinear ICS (1) requires more nuanced consideration than conventional ordinary differential equation. This requirement inherently elevates an additional layer of complexity in determining appropriate coordinate transformations. On the other hand, we observed that Assumption 3 is always satisfied when adopting linear transformations of the form $z = Tx$, where T is a nonsingular $n \times n$ matrix.

In the following, the distribution and the co-distribution are used to obtain two kinds of local decompositions for nonlinear ICS (1).

Theorem 1. (a) Let Δ be a nonsingular involutive distribution of dimension d , and assume that Δ is invariant under $f, g_1, \dots, g_m, \varphi^\dagger$ and contains $\text{span}\{g_1, \dots, g_m\}$. Then, for each point $x_0 \in U$, it is possible to find a neighborhood U_0 of x_0 and a coordinates transformation $z = \Psi(x) = \text{col}(\psi_1(x), \dots, \psi_n(x))$ defined on U_0 . If Assumptions 2 and 3 are satisfied, then the nonlinear ICS (1) has the following controllability structure:

$$\begin{cases} \dot{\xi}_1 = f_1(\xi_1, \xi_2) + \sum_{i=1}^m g_{1i}(\xi_1, \xi_2)u_i, & t \neq \tau_k, \\ \dot{\xi}_2 = f_2(\xi_2), & t \neq \tau_k, \\ \xi_1(\tau_k^+) = \varphi_1(\xi_1(\tau_k), \xi_2(\tau_k)), & t = \tau_k, \\ \xi_2(\tau_k^+) = \varphi_2(\xi_2(\tau_k)), & t = \tau_k, \\ y_j = h_{1j}(\xi_1, \xi_2), & 1 \leq j \leq p. \end{cases} \quad (8)$$

where $\xi_1 = (z_1, \dots, z_d)$, $\xi_2 = (z_{d+1}, \dots, z_n)$, and $f_1, f_2, \varphi_1, \varphi_2, h_{1j}, g_{1i}$ are smooth functions.

(b) Let Δ be a nonsingular involutive distribution of dimension d , and assume that that Δ is invariant under $f, g_1, \dots, g_m, \varphi^\dagger$ and $\Delta^\perp$ contains $\text{span}\{dh_1, \dots, dh_p\}$. Then, for each point $x_0 \in U$, it is possible to find a neighborhood U_0 of x_0 and a coordinates transformation $z = \Psi(x) = \text{col}(\psi_1(x), \dots, \psi_n(x))$ defined on U_0 . If Assumptions 2 and 3 are satisfied, then the nonlinear ICS (1) has the following observability structure:

$$\begin{cases} \dot{\xi}_1 = f_1(\xi_1, \xi_2) + \sum_{i=1}^m g_{1i}(\xi_1, \xi_2)u_i, & t \neq \tau_k, \\ \dot{\xi}_2 = f_2(\xi_2) + \sum_{i=1}^m g_{2i}(\xi_2)u_i, & t \neq \tau_k, \\ \xi_1(\tau_k^+) = \varphi_1(\xi_1(\tau_k), \xi_2(\tau_k)), & t = \tau_k, \\ \xi_2(\tau_k^+) = \varphi_2(\xi_2(\tau_k)), & t = \tau_k, \\ y_j = h_{1j}(\xi_2), & 1 \leq j \leq p. \end{cases} \quad (9)$$

where $\xi_1 = \text{col}(z_1, \dots, z_d)$, $\xi_2 = \text{col}(z_{d+1}, \dots, z_n)$, and $f_1, f_2, \varphi_1, \varphi_2, h_{1j}, g_{1i}, g_{2i}$ are smooth functions.

Proof. Since Δ is nonsingular involutive, from Lemma 1, we have the distribution Δ is complete integrable. Thus, there exists a local smooth coordinates transformation $z = \Psi(x)$ defined on U_0 (locally around x_0) with the property that

$$\text{span}\{d\psi_{d+1}, \dots, d\psi_n\} = \Delta^\perp.$$

Since $\varphi(U_0) \subseteq U_0$ and (3) is satisfied, we could do a local coordinates transformation of the ICS (1). Furthermore, in the new coordinates, $\forall \alpha \in \Delta$ has a representation of the form

$$\bar{\alpha}(z) = \text{col}(\bar{\alpha}_1(z), \dots, \bar{\alpha}_d(z), 0, \dots, 0) \in \Delta, \quad (10)$$

and $\forall \beta \in \Delta^\perp$ has a representation of the form

$$\bar{\beta}(z) = \text{col}(0, \dots, 0, \bar{\beta}_{d+1}(z), \dots, \bar{\beta}_n(z)) \in \Delta^\perp. \quad (11)$$

Note that, $\bar{\alpha}$ and $\bar{\beta}$ denote the representation of α and β in the new coordinates. They have simple representation due to the complete integrability of Δ .

Then, consider the following vector field

$$\bar{\alpha}^q(z) = \text{col}(0, 0, \dots, \underset{q\text{-th}}{1}, \dots, 0). \quad (12)$$

Recall that, in the coordinates just chosen, every vector field of Δ is characterized by the property that the last $n - d$ components are vanishing. Thus, if $1 \leq i \leq d$, then $\alpha^q, \bar{\alpha}^q \in \Delta$, where α^q denotes the representation of $\bar{\alpha}^q$ in the original coordinates.

For the impulsive dynamics of (1), denote $\bar{\varphi}$ and $\bar{\varphi}^\dagger$ the representation of φ and $\varphi^\dagger$ in the new coordinates, respectively. By (6), we have

$\bar{\varphi}^\dagger(z) = [\frac{\partial \Psi}{\partial x} \varphi^\dagger(x)]|_{x=\Psi^{-1}(z)}$ and

$$\begin{aligned} [\bar{\varphi}^\dagger, \bar{\alpha}^q] &= \frac{\partial \bar{\alpha}^q}{\partial z} \bar{\varphi}^\dagger - \frac{\partial \bar{\varphi}^\dagger}{\partial z} \bar{\alpha}^q \\ &= 0 - \begin{bmatrix} \frac{\partial \bar{\varphi}_1^\dagger}{\partial z_1} & \frac{\partial \bar{\varphi}_1^\dagger}{\partial z_2} & \dots & \frac{\partial \bar{\varphi}_1^\dagger}{\partial z_n} \\ \frac{\partial \bar{\varphi}_2^\dagger}{\partial z_1} & \frac{\partial \bar{\varphi}_2^\dagger}{\partial z_2} & \dots & \frac{\partial \bar{\varphi}_2^\dagger}{\partial z_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \bar{\varphi}_n^\dagger}{\partial z_1} & \frac{\partial \bar{\varphi}_n^\dagger}{\partial z_2} & \dots & \frac{\partial \bar{\varphi}_n^\dagger}{\partial z_n} \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \\ &= -\text{col}(\frac{\partial \bar{\varphi}_1^\dagger}{\partial z_q}, \frac{\partial \bar{\varphi}_2^\dagger}{\partial z_q}, \dots, \frac{\partial \bar{\varphi}_n^\dagger}{\partial z_q}), \end{aligned} \quad (13)$$

where $\bar{\varphi}_l^\dagger, 1 \leq l \leq n$, is the l th component of $\bar{\varphi}^\dagger$. Since Δ is invariant under $\varphi^\dagger$, we have $[\varphi^\dagger, \alpha^q] \in \Delta$. It implies that $[\bar{\varphi}^\dagger, \bar{\alpha}^q] \in \Delta$. Thus, for $d+1 \leq k \leq n, 1 \leq q \leq d$, we obtain that $\frac{\partial \bar{\varphi}_k^\dagger}{\partial z_q} = 0$. From (7), we have $\bar{\varphi}(z) = \bar{\varphi}^\dagger(z) + z$. Therefore, we obtain that $\frac{\partial \bar{\varphi}_k}{\partial z_q} = 0$, for $d+1 \leq k \leq n, 1 \leq q \leq d$.

For the continuous dynamics of (1), denote $\bar{f}$ the representation of f in the new coordinates, and denote $\bar{g}_i, i = 1, \dots, m$ the representation of g_i in the new coordinates. By (1) and $z = \Psi(x)$, we obtain $\bar{f} = \frac{\partial \Psi}{\partial x} f, \bar{g}_i = \frac{\partial \Psi}{\partial x} g_i$. Since Δ is invariant under $f, g_1, \dots, g_m$, we have $[\bar{f}, \bar{\alpha}^q], [\bar{g}_i, \bar{\alpha}^q]$ belongs to Δ . Similarly, there are $\frac{\partial \bar{f}_k}{\partial z_q} = \frac{\partial \bar{g}_{ik}}{\partial z_q} = 0$, where $\bar{f}_k$ is the k th component of $\bar{f}$, $\bar{g}_{ik}$ is the k th component of $\bar{g}_i, d+1 \leq k \leq n, 1 \leq q \leq d$.

Hence, the nonlinear ICS (1) is transformed into the following form in the new coordinates:

$$\begin{cases} \dot{\xi}_1 = f_1(\xi_1, \xi_2) + \sum_{i=1}^m g_{1i}(\xi_1, \xi_2)u_i, & t \neq \tau_k, \\ \dot{\xi}_2 = f_2(\xi_2) + \sum_{i=1}^m g_{2i}(\xi_2)u_i, & t \neq \tau_k, \\ \xi_1(\tau_k^+) = \varphi_1(\xi_1(\tau_k), \xi_2(\tau_k)), & t = \tau_k, \\ \xi_2(\tau_k^+) = \varphi_2(\xi_2(\tau_k)), & t = \tau_k, \\ y_j = h_{1j}(\xi_1, \xi_2), & 1 \leq j \leq p, \end{cases} \quad (14)$$

where f, g_i, φ, h_j are now represented in the new coordinates by $\text{col}(f_1, f_2), \text{col}(g_{1i}, g_{2i}), \text{col}(\varphi_1, \varphi_2), h_{1j}$, respectively.

If Δ contains $\text{span}\{g_1, \dots, g_m\}$. Since in the new coordinates, g_i is represented by $\text{col}(g_{1i}(z), g_{2i}(z)) = [\frac{\partial \Psi}{\partial x} g_i(x)]|_{x=\Psi^{-1}(z)}$ and $g_i \in \Delta$, we have $\text{col}(g_{1i}(z), g_{2i}(z))$ has a representation of the form by (10):

$$\text{col}(g_{1i}(z), g_{2i}(z)) = \text{col}(g_{i1}(z), \dots, g_{id}(z), 0, \dots, 0). \quad (15)$$

Combining (14) and (15), the proof of (a) is complete.

If $\Delta^\perp$ contains $\text{span}\{dh_1, \dots, dh_p\}$, then the co-vector fields $dh_1, \dots, dh_p$ belong to $\Delta^\perp$. Hence, in the new coordinates, dh_{1j} have the form of (11). Note that $dh_{1j} = (\frac{\partial h_{1j}}{\partial z_1} \frac{\partial h_{1j}}{\partial z_2} \dots \frac{\partial h_{1j}}{\partial z_n})$. Thus, we obtain that $\frac{\partial h_{1j}}{\partial z_k} = 0, 1 \leq j \leq p, 1 \leq k \leq d$. In this case, we obtain $y_j = h_{1j}(\xi_2)$. The proof of (b) is complete. ■

By this theorem, Kalman controllable decomposition and Kalman observable decomposition are generalized to the nonlinear ICS (1). In the following sections, they are frequently utilized.

Next, consider the following linear ICS:

$$\begin{cases} \dot{x}(t) = (A + BF)x(t), & t \neq \tau_k, \\ x(\tau_k^+) = Jx(\tau_k), & t = \tau_k, \\ y(t) = Cx(t), \end{cases} \quad (16)$$

where A, B, F, J, C are constant matrices having appropriate dimensions. It follows from Theorem 1(b) that we have the special observability form for (16).

Corollary 1. For the linear ICS (16), suppose the statement in Theorem 1(b) holds. Then the linear ICS (16) has the following observability

structure:

$$\begin{cases} \dot{z}(t) = \begin{bmatrix} \bar{A}_{11} + \bar{B}_1 \bar{F}_1 & \bar{A}_{12} + \bar{B}_1 \bar{F}_2 \\ 0 & \bar{A}_{22} + \bar{B}_2 \bar{F}_2 \end{bmatrix} z(t) & t \neq \tau_k, \\ z(\tau_k) = \begin{bmatrix} \bar{J}_{11} & \bar{J}_{12} \\ 0 & \bar{J}_{22} \end{bmatrix} z(\tau_k^-), & t = \tau_k, \\ y(t) = [0 \quad \bar{C}_2] z(t), \end{cases} \quad (17)$$

where $\bar{A}_{ij}, \bar{B}_i, \bar{J}_{ij}, \bar{F}_i, \bar{C}_2, i, j \in \{1, 2\}, (\bar{A}_{21} = \bar{J}_{21} = 0)$ are constant matrices having appropriate dimensions.

Remark 1. Considering the system's form after the coordinates transformation, the observability form (17) coincides with the compensated linear impulsive system Σ_F without disturbance in [32].

3.2. Accessibility

In this subsection, we focus our attention on accessibility properties based on the geometric control theory. We begin with the following definitions inspired by [33].

Definition 3. For nonlinear ICS (1) with a given $\mathcal{T}$, let $\mathcal{R}^V(x_0, t_f, \mathcal{T})$ be the reachable set from x_0 at a specific time $t_f > 0$, following trajectories which remain for $t \leq t_f$ in the neighborhood V of x_0 , i.e., $\mathcal{R}^V(x_0, t_f, \mathcal{T}) = \{x \in U : \text{there exists admissible control such that the evolution of nonlinear ICS (1) for } x(0) = x_0 \text{ satisfies } x(t) \in V, 0 < t \leq t_f \text{ and } x(t_f) = x\}$. Given $T \geq t_f > 0$, Denote

$$\mathcal{R}^V(x_0, \mathcal{T}) = \bigcup_{t_f \in (0, T]} \mathcal{R}^V(x_0, t_f, \mathcal{T}). \quad (18)$$

Then, consider $V = U$, we denote $\mathcal{R}(x_0, \mathcal{T}) := \mathcal{R}^U(x_0, \mathcal{T})$ and $\mathcal{R}(x_0, t_f, \mathcal{T}) := \mathcal{R}^U(x_0, t_f, \mathcal{T})$. Note that, $\mathcal{R}(x_0, t_f, \mathcal{T})$ means the reachable set of $x_0 \in U$ at time t_f .

Definition 4. The nonlinear ICS (1) is said to be

- (i) *accessible* from $x_0 \in U$, if $\mathcal{R}(x_0, \mathcal{T})$ contains a nonempty open set of U for the given $\mathcal{T}$ and $T > \tau_1$. If this holds for every $x_0 \in U$, then the system (1) is said to be accessible.
- (ii) *strongly accessible* from $x_0 \in U$, if $\mathcal{R}(x_0, t_f, \mathcal{T})$ contains a nonempty open set of U for the given $\mathcal{T}$ and for any $t_f > \tau_1$. If this holds for every $x_0 \in U$, then the system (1) is said to be strongly accessible.

Next, we introduce the accessibility distribution R and strong accessibility distribution P : R is the smallest distribution invariant under $f, \varphi^\dagger, g_1, \dots, g_m$ which contains $\text{span}\{f, g_1, \dots, g_m\}$, and P is the smallest distribution invariant under $f, \varphi^\dagger, g_1, \dots, g_m$ which contains $\text{span}\{g_1, \dots, g_m\}$. To simplify the notation, we set

$$\begin{aligned} R(x) &= \langle f, \varphi^\dagger, g_1, \dots, g_m | \text{span}\{f, g_1, \dots, g_m\} \rangle(x), \\ P(x) &= \langle f, \varphi^\dagger, g_1, \dots, g_m | \text{span}\{g_1, \dots, g_m\} \rangle(x). \end{aligned} \quad (19)$$

Similar to Lemma 1.8.5 and Lemma 1.8.7 in [1], we have the following lemma.

Lemma 2. Here are some properties regarding R and P .

- (i) If R is nonsingular, then R is involutive. Similarly, If P is nonsingular, then P is involutive. If P and $P + \text{span}\{f\}$ are nonsingular, then $P + \text{span}\{f\} = R$.
- (ii) $\dim R \leq \dim P + 1$.

In the following, we show the method for finding P .

Lemma 3. Set

$$\begin{aligned} P_0 &= \text{span}\{g_1, \dots, g_m\} \\ P_i &= P_{i-1} + [f, P_{i-1}] + [\varphi^\dagger, P_{i-1}] + \sum_{j=1}^m [g_j, P_{i-1}], \quad i \geq 1. \end{aligned} \quad (20)$$

If there exists an integer i^* such that $P_{i^*} = P_{i^*+1}$, then

$$P = P_{i^*} = \langle f, \varphi^\dagger, g_1, \dots, g_m | \text{span}\{g_1, \dots, g_m\} \rangle. \quad (21)$$

Lemma 3 can be obtained from Lemma 1.8.2 in [1]. Note that, for the nonlinear ICS (1), the term $[\varphi^\dagger, P_{i-1}]$ is added in (20) to guarantee the invariance of P_i with respect to $\varphi^\dagger$. Similarly, we can find R by a process by setting $R_0 = \text{span}\{f, g_1, \dots, g_m\}$ and replacing P_i, P_{i-1} with R_i, R_{i-1} in (20).

Next we apply Lemma 3 on a linear ICS. Consider $f(x) = A_C x$, $[g_1, \dots, g_m](x) = B_C$, $\varphi(x) = A_I x$, $[h_1, \dots, h_p](x) = C_C x$ in (1), which is the linear ICS in [15]:

$$\begin{cases} \dot{x}(t) = A_C x(t) + B_C u(t), & t \neq \tau_k, \\ x(\tau_k^+) = A_I x(\tau_k), & t = \tau_k, \\ y(t) = C_C x(t), \end{cases} \quad (22)$$

where A_C, B_C, A_I , and C_C are matrices having appropriate dimensions, $u(t)$ is the corresponding control input.

For linear ICS (22), let $B = \text{Im}(B_C)$, and denote $b_1, \dots, b_m$ the columns of B_C . We have

$$P_0 = B,$$

$$P_i = P_{i-1} + [A_C x, P_{i-1}] + [A_I x - x, P_{i-1}] + \sum_{j=1}^m [b_j, P_{i-1}], i \geq 1.$$

Since

$$[A_C x, b_j] = -\frac{\partial A_C x}{\partial x} b_j = -A_C b_j,$$

$$[A_I x - x, b_j] = -\frac{\partial (A_I x - x)}{\partial x} b_j = -A_I b_j + b_j$$

and $[b_l, b_j] = 0$, where $1 \leq l, j \leq m$, we have

$$\begin{aligned} P_1 &= \text{span}\{b_1, \dots, b_m, A_C b_1, \dots, A_C b_m, A_I b_1, \dots, A_I b_m\} \\ &= B + A_C B + A_I B. \end{aligned} \quad (23)$$

Continuing in the same way, we deduce that

$$P_i = P_{i-1} + A_C P_{i-1} + A_I P_{i-1}. \quad (24)$$

Remark 2. For the linear ICS (22), Lemma 3 is the same as Algorithm 4.3 in [15].

Next, we will use the structural decomposition in Theorem 1(a) and distributions defined above to present some accessibility results of nonlinear ICS (1).

Theorem 2. Suppose R defined by (19) is nonsingular with dimension r . Then, for each $x_0 \in U$, it is possible to find a neighborhood U° of x_0 and a coordinates transformation $z = \Psi(x) = \text{col}(\psi_1(x), \dots, \psi_n(x))$ defined on U° . If φ is a diffeomorphism from U° into its image such that Assumptions 2 and 3 are satisfied, then we have following properties:

(a) $R^{U^\circ}(x_0, \mathcal{T})$ is a subset of $\bigcup_{0 \leq l \leq l_0} S_{x_0, \mathcal{T}, l}$, where

$$S_{x_0, \mathcal{T}, l} = \{x \in U^\circ \mid \psi_i(x) = \psi_i(\varphi^l(x_0))\}$$

with $r+1 \leq i \leq n$, $\varphi^0(x_0) = x_0$, $\varphi^l(x_0) = \underbrace{\varphi \circ \varphi \circ \dots \circ \varphi}_{l\text{-times}}(x_0)$, l_0 being

the number of impulses within the time interval $(0, T)$ for a given $\mathcal{T}$.

(b) If there are $l_0 \geq 1$ impulses within the time interval $(0, T)$, then $R^{U^\circ}(x_0, \mathcal{T})$ contains an open subset of $S_{x_0, \mathcal{T}, l_0}$.

Proof. We first prove the statement (a). Since R is nonsingular, we have R is involutive by Lemma 2. Combining the proof in Theorem 1, we can obtain the following structural decomposition for nonlinear ICS

(1) under the distribution R :

$$\begin{cases} \dot{\xi}_1 = f_1(\xi_1, \xi_2) + \sum_{i=1}^m g_{1i}(\xi_1, \xi_2) u_i, & t \neq \tau_k, \\ \dot{\xi}_2 = 0, & t \neq \tau_k, \\ \xi_1(\tau_k) = \varphi_1(\xi_1(\tau_k^-), \xi_2(\tau_k^-)), & t = \tau_k, \\ \xi_2(\tau_k) = \varphi_2(\xi_2(\tau_k^-)), & t = \tau_k, \\ y_j = h_{1j}(\xi_1, \xi_2), & 1 \leq j \leq p. \end{cases} \quad (25)$$

where $\xi_1 = \text{col}(z_1, \dots, z_r)$, $\xi_2 = \text{col}(z_{r+1}, \dots, z_n)$. From (25), we have $z_j(x(t)) = \psi_j(x(t)) = \psi_j(\varphi^l(x_0))$, $j = r+1, \dots, n$, $\tau_l \leq t < T$, $0 \leq l$ and setting $\tau_0 = 0$. Hence, the proof of (a) is complete.

Next, we proceed to the proof of (b). Given $\tau_1, \dots, \tau_{l_0}$ the impulse time within the time interval $(0, T)$. We then pick time variables $t_1, \dots, t_r \in (0, \epsilon)$, where $t_1 > \tau_{l_0}$ and ϵ is a constant.

From supposition, we have $\dim R(x) = r, \forall x \in U^\circ$. If for any $\theta = f + \sum_{i=1}^m g_i u_i \in R$ there is $\theta(x_0) = 0$, then it implies that $\dim R(x_0) = 0$ which contradicts the supposition $\dim R(x) = r$. So, there exists a $\theta_1 = f + \sum_{i=1}^m g_i u_i^1 \in R$, such that $\theta_1(x) \neq 0, \forall x \in U_1 \subseteq U^\circ$. Then for $\epsilon > 0$ and $\tau_{l_0} < t_1 < \epsilon$, we have $\Phi_{t_1}^{\theta_1}$, which denotes the corresponding flow without impulse. $\Phi_{t_1}^{\theta_1}$ is an one-dimensional manifold, denoted as L_1 . Since φ is a diffeomorphism, we have $\varphi(L_1)$ is also an one-dimensional manifold, denoted as $L_{1,2} \subseteq U^\circ$. Set $x_{p1} = \varphi(\Phi_{t_1}^{\theta_1}(x_0)) \in L_{1,2}$. Then there exists a $\theta_{1,1} = f + \sum_{i=1}^m g_i u_i^{1,1} \in R$, such that $\theta_{1,1}(x_{p1}) \in T_{x_{p1}} L_{1,2}$. Denote $L_{2,1} = \Phi_{t_1-\tau_1}^{\theta_{1,1}}(x_{p1})$ that is an one-dimensional manifold. Then, we have $L_{2,2} = \varphi(L_{2,1})$ is an one-dimensional manifold. Set $x_{p2} = \varphi(\Phi_{t_2-\tau_1}^{\theta_{1,1}}(x_{p1})) \in L_{2,2}$. Then there exists a $\theta_{1,2} = f + \sum_{i=1}^m g_i u_i^{1,2} \in R$, such that $\theta_{1,2}(x_{p2}) \in T_{x_{p2}} L_{2,2}$. Denote $L_{3,1} = \Phi_{t_1-\tau_2}^{\theta_{1,2}}(x_{p2})$ that is an one-dimensional manifold. Then, we have $L_{3,2} = \varphi(L_{3,1})$ is an one-dimensional manifold. Set $x_{p3} = \varphi(\Phi_{t_3-\tau_2}^{\theta_{1,2}}(x_{p2})) \in L_{3,2}$.

This procedure can be iterated and finally we have that there exists a $\theta_{1,l_0} = f + \sum_{i=1}^m g_i u_i^{1,l_0} \in R$, such that $\theta_{1,l_0}(x_{pl_0}) \in T_{x_{pl_0}} L_{l_0,2}$. Denote $\bar{L}_1^* = \Phi_{t_1-\tau_{l_0}}^{\theta_{1,l_0}}(x_{pl_0})$ that is an one-dimensional manifold.

Due to $\dim \bar{L}_1^* = 1$, there exists $\tau_{l_0} < s_1^1 < \epsilon$ and $\theta_2 = f + \sum_{i=1}^m g_i u_i^2 \in R$ such that $\theta_2(x_1) \notin T_{x_1}(\bar{L}_1^*)$, where $x_1 = \Phi_{s_1^1-\tau_{l_0}}^{\theta_{1,l_0}}(x_{pl_0})$. Otherwise, $\dim R(x) = 1, \forall x \in \bar{L}_1^*$. Consider the following mapping:

$$\pi_2(t_2, t_1) = \Phi_{t_2}^{\theta_2} \circ \Phi_{t_1-\tau_{l_0}}^{\theta_{1,l_0}}(x_{pl_0}). \quad (26)$$

Then, we have $\frac{\partial \pi_2(t_2, s_1^1)}{\partial t_2} \big|_{t_2=0} = \theta_2(x_1)$ and $\frac{\partial \pi_2(0, t_1)}{\partial t_1} \big|_{t_1=s_1^1} \in T_{x_1} \bar{L}_1^*$. The latter can be interpreted as: the time derivative of a curve on a manifold (its velocity vector) is contained within the tangent space of the manifold at the point where the curve is located. Therefore $\pi_2(t_2, t_1)$ has rank equal to 2 at the point $(0, s_1^1)$. Hence $\pi_2(t_2, t_1)$ is a local diffeomorphism from $0 < t_2 < \epsilon, s_1^1 < t_1 < \epsilon$ into its image. We denote this image as L_2 , which is a two-dimensional manifold.

Similarly, there exist $0 < s_2^2 < \epsilon$ and $s_1^1 < s_1^2 < \epsilon$ such that $\theta_3(x_2) \notin T_{x_2}(L_2)$, where $x_2 = \Phi_{s_2^2}^{\theta_2} \circ \Phi_{s_1^2-\tau_{l_0}}^{\theta_{1,l_0}}(x_{pl_0})$. Then we consider the following mapping:

$$\pi_3(t_3, t_2, t_1) = \Phi_{t_3}^{\theta_3} \circ \Phi_{t_2}^{\theta_2} \circ \Phi_{t_1-\tau_{l_0}}^{\theta_{1,l_0}}(x_{pl_0}). \quad (27)$$

We denote the image of $\pi_3(t_3, t_2, t_1)$ as L_3 , which is a three-dimensional manifold. Iterating this procedure, then we can find $\theta_4, \theta_5, \dots, \theta_r \in R$ and define the following mapping:

$$\begin{aligned} \pi_r : (0, \epsilon) \times (s_{r-1}^{r-1}, \epsilon) \times \dots \times (s_1^{r-1}, \epsilon) &\rightarrow U^\circ \\ (t_r, \dots, t_1) &\mapsto \Phi_{t_r}^{\theta_r} \circ \dots \circ \Phi_{t_1-\kappa_1}^{\theta_1} \circ \varphi \circ \Phi_{\kappa_1}^{\theta_1}(x_0). \end{aligned} \quad (28)$$

We set an open set V_r as

$$V_r = (0, \epsilon) \times (s_{r-1}^{r-1}, \epsilon) \times \dots \times (s_1^{r-1}, \epsilon). \quad (29)$$

It follows that π_r is a local diffeomorphism from V_r into its image $\pi_r(V_r)$ which is r -dimensional manifold and is denoted as M_{r1} .

Observe that under the following control

$$\begin{cases} u_i(t) = u_i^1 & t \in [0, \tau_1], \\ u_i(t) = u_i^{1,1} & t \in (\tau_1, \tau_2], \\ u_i(t) = u_i^{1,2} & t \in (\tau_2, \tau_3], \\ \dots \\ u_i(t) = u_i^{1,l_0} & t \in (\tau_{l_0}, t_1], \\ u_i(t) = u_i^2 & t \in (t_1, t_1 + t_2], \\ \dots \\ u_i(t) = u_i^r & t \in (t_1 + \dots + t_{r-1}, t_1 + \dots + t_r], \end{cases} \quad (30)$$

a point $\bar{x} = \pi_r(t_r, \dots, t_2, t_1) \in M_r$ can be reached, starting from the initial state x_0 and affected by l_0 impulses. Therefore, $M_r \subseteq \mathcal{R}^{U^\circ}(x_0, \mathcal{T})$. We know from (a) that M_r is contained in:

$$S_{x_0, \mathcal{T}, l_0} = \{x \in U^\circ \mid \psi_i(x) = \psi_i(\varphi^{l_0}(x_0))\}, \quad (31)$$

where $r+1 \leq i \leq n$. Since $\dim M_r = r$ and M_r is the image under π_r of the open set V_r , we have M_r is an open set of $S_{x_0, \mathcal{T}, l_0}$. Therefore, $\mathcal{R}^{U^\circ}(x_0, \mathcal{T})$ contains an open subset of $S_{x_0, \mathcal{T}, l_0}$. The proof of (b) is complete. ■

Remark 3. In this theorem, we provide a geometric characterization of $\mathcal{R}^{U^\circ}(x_0, \mathcal{T})$. If the nonlinear ICS (1) is impulse-free, then we can obtain Theorem 1.8.9 in [1] from this theorem.

It is noted that, if $\dim R(x) = n$, we have ICS (1) itself is a controllable form. Hence, there is no need to do a local coordinates transformation for it. Besides, we have $\bigcup_{1 \leq i \leq l_0} M_{r1}$, which in the proof of Theorem 2, is an open set of U . And there is $\mathcal{R}^{U^\circ}(x_0, \mathcal{T}) \subseteq \mathcal{R}(x_0, \mathcal{T})$. Therefore, $\mathcal{R}(x_0, \mathcal{T})$ contains an open set of U . Then we have the following corollary.

Corollary 2. Suppose R defined by (19) is nonsingular and φ is a diffeomorphism from U into its image. If $\dim R(x) = n$ for all $x \in U$, then the ICS (1) is accessible.

Definition 5. The condition $\dim R(x_0) = n$ is called the *accessibility rank condition* at x_0 . If $\dim R(x_0) = n$ holds for any $x_0 \in U$, then the system (1) is said to satisfy the accessibility rank condition.

Regarding strong accessibility, we have the following results.

Theorem 3. Suppose P defined by (19) and $P + \text{span}\{f\}$ are nonsingular. Let p denote the dimension of P . Then, for each $x_0 \in U$, it is possible to find a neighborhood U° of x_0 and a coordinates transformation $z = \Psi(x) = \text{col}(\psi_1(x), \dots, \psi_n(x))$ defined on U° . If φ is a diffeomorphism from U° into its image such that Assumptions 2 and 3 are satisfied, then for $0 < \tau_1 < \dots < \tau_{l_0} < t_f$, $1 \leq l_0$ we have following properties:

- (a) $\mathcal{R}^{U^\circ}(x_0, t_f, \mathcal{T})$ is a subset of $S_{x_0, t_f, \mathcal{T}} = \{x \in U^\circ \mid \psi_i(x) = \psi_i(\varphi_{t_f - \tau_{l_0}}^{f} \circ \varphi_{\tau_{l_0} - \tau_{l_0-1}}^{f} \circ \dots \circ \varphi_{\tau_1}^{f}(x_0))\}$, where $p+1 \leq i \leq n$, $\tau_0 = 0$.
- (b) $\mathcal{R}^{U^\circ}(x_0, t_f, \mathcal{T})$ contains an open subset of $S_{x_0, t_f, \mathcal{T}}$.

Proof. The proof for (a) is derived from Theorem 1(a). Now, let us focus on proving (b), which can be reduced to the proof of Theorem 2. From Lemma 2, we have R is nonsingular, and the dimension of R denoted by r , is equal to p or $p+1$.

Suppose $r = p+1$ first. We then pick time variables $t_1, \dots, t_r \in (0, \epsilon)$, where $t_1 > \tau_{l_0}$ and ϵ is a constant. Given any real number $t_f \in (\tau_{l_0}, \epsilon)$, consider the set:

$$V_r^{t_f} = \{(t_r, \dots, t_1) \in V_r \mid t_1 + \dots + t_r = t_f\}, \quad (32)$$

where V_r is described by (29). Clearly, by (28), the image of $\pi_r(V_r^{t_f})$ denoted as $M_r^{t_f}$, consists of points reachable at time t_f from x_0 with

l_0 impulses. Therefore, $M_r^{t_f} \subseteq \mathcal{R}^{U^\circ}(x_0, t_f, \mathcal{T})$, and we know from (a) in this theorem that $M_r^{t_f} \subseteq S_{x_0, t_f, \mathcal{T}}$. Similar to the proof of Theorem 2, we have $M_r^{t_f}$ is an open subset of $S_{x_0, t_f, \mathcal{T}}$.

Next, suppose $r = p$, namely, $R = P$. Augment the nonlinear ICS (1) by the equation $i = 1$, t being the time variable, so that we have the augmented system:

$$\begin{cases} \dot{x}(t) = f(x(t)) + \sum_{i=1}^m g_i(x(t))u_i(t), & t \neq \tau_k, \\ i = 1, & t \neq \tau_k, \\ x(\tau_k^+) = \varphi(x(\tau_k)), & t = \tau_k, \\ t(\tau_k^+) = t(\tau_k), & t = \tau_k, \\ y_j(t) = h_j(x(t)), & 1 \leq j \leq p, \end{cases} \quad (33)$$

defined on $\bar{U} = U \times \mathbb{R}$ with $\bar{x} = (x, t)$, $\bar{f}(\bar{x}) = \begin{pmatrix} f(x) \\ 1 \end{pmatrix}$, $\bar{g}_i(\bar{x}) = \begin{pmatrix} g_i(x) \\ 0 \end{pmatrix}$, $\bar{\varphi}^+(\bar{x}) = \begin{pmatrix} \varphi(x) - x \\ 0 \end{pmatrix}$. Denote $\bar{R} = \langle \bar{f}, \bar{\varphi}^+, \bar{g}_1, \dots, \bar{g}_m \mid \text{span}\{\bar{f}, \bar{g}_1, \dots, \bar{g}_m\} \rangle$, we have $\dim \bar{R} = p+1$. By Theorem 2, $\mathcal{R}^{U^\circ}((x_0, t_0), \mathcal{T})$ contains an open subset of $M_r \times (\tau_{l_0}, t_f) \subseteq U \times \mathbb{R}$. Hence, there exists an interval (a, b) , with $0 < \tau_{l_0} < a < b \leq t_f$, such that

$$M_r \times (a, b) \subseteq \mathcal{R}^{U^\circ}((x_0, t_0), \mathcal{T}).$$

Let $\Xi \in (a, b)$. Since $M_r \times \{\Xi\} \subseteq \mathcal{R}^{U^\circ}((x_0, t_0), \mathcal{T})$, we have $M_r \subseteq \mathcal{R}^{U^\circ}(x_0, \Xi, \mathcal{T})$. Choose any $\theta \in P$, the mapping $G : x \mapsto \Phi_{t_f - \Xi}^\theta(x)$ maps M_r onto an open subset $\bar{M}_r \subseteq \mathcal{R}^{U^\circ}(x_0, t_f, \mathcal{T})$. Note that $\bar{M}_r \subseteq S_{x_0, t_f, \mathcal{T}}$, the proof of (b) is complete. ■

Remark 4. In this theorem, we provide a geometric characterization of $\mathcal{R}^{U^\circ}(x_0, t_f, \mathcal{T})$. If the nonlinear ICS (1) is impulse-free, then we can obtain Theorem 1.8.10 in [1] from this theorem.

Similar to Corollary 2, we have the following corollary.

Corollary 3. Suppose P defined by (19) is nonsingular and φ is a diffeomorphism from U into its image. If $\dim P(x) = n$ for all $x \in U$, then the system (1) is strongly accessible.

Definition 6. The condition $\dim P(x_0) = n$ is called the *strong accessibility rank condition* at x_0 . If $\dim P(x_0) = n$ holds for any $x_0 \in U$, then the system (1) is said to satisfy the strong accessibility rank condition.

As an application, we consider the following system:

$$\begin{cases} \dot{x}(t) = A_C x(t) + B_C u(t), & t \neq \tau_k, \\ x(\tau_k^+) = d x(\tau_k), & t = \tau_k, \\ y(t) = C_C x(t), \end{cases} \quad (34)$$

where $d \neq 0$ is a constant, A_C, B_C , and C_C are matrices having appropriate dimensions. The controllability for (34) has been studied in [14].

Definition 7 ([14]). Linear ICS (34) is called *controllable* on $[0, t_f]$ if given any initial state $x_0 \in U$, there exists a piecewise continuous input $u(t)$ such that the corresponding solution of (34) satisfies $x(t_f) = 0$.

Proposition 1. The following statements are equivalent:

- (i) Linear ICS (34) is controllable on $[0, t_f]$.
- (ii) Linear ICS (34) satisfies the strong accessibility rank condition.
- (iii) Linear ICS (34) is strongly accessible.

Proof. It can be calculated from Lemma 3 that the strong accessibility rank condition for (34) is

$$\text{rank}[B_C, A_C B_C, \dots, A_C^{n-1} B_C] = n. \quad (35)$$

Thus, (i) $\Leftrightarrow$ (ii) is the result from Theorem 3.2 in [14]. Then, it follows from Corollary 2 that (ii) $\Rightarrow$ (iii).

For (iii) $\Rightarrow$ (ii), we consider the following impulse-free system corresponding to linear ICS (34) first:

$$\begin{cases} \dot{p}(t) = A_C p(t) + B_C \bar{u}(t), \\ \bar{y}(t) = C_C p(t). \end{cases} \quad (36)$$

$\bar{u}(t)$ is a piecewise continuous input. We claim that if the linear ICS (34) is strongly accessible, then the corresponding impulse-free system (36) is strongly accessible. For, denote $\mathcal{R}(x_0, t, \mathcal{T})$ as the reachable set of the linear ICS (34) from x_0 at time t , and denote $\mathcal{R}(p_0, t)$ as the reachable set of the impulse-free system (36) from p_0 at time t . For $t \in (\tau_1, \tau_2)$ and any $x_0 \in U$, we have the general solution of system (34) is given by

$$x(t) = d e^{A_C(t-\tau_1)} x(\tau_1) + \int_{\tau_1}^t e^{A_C(t-s)} B u(s) ds. \quad (37)$$

Consider $\bar{u}(t) = u(t)$ when $t \in (0, \tau_1]$ and $\bar{u}(t) = \frac{1}{d} u(t)$ when $t \in (\tau_1, t]$. Then we have $x(\tau_1) = p(\tau_1)$ for $p_0 = x_0$. On the other hand, for $t \in (\tau_1, \tau_2)$, we have the general solution of system (36) is given by

$$\begin{aligned} p(t) &= e^{A_C(t-\tau_1)} p(\tau_1) + \int_{\tau_1}^t e^{A_C(t-s)} B \bar{u}(s) ds \\ &= \frac{1}{d} (d e^{A_C(t-\tau_1)} p(\tau_1) + \int_{\tau_1}^t e^{A_C(t-s)} B u(s) ds). \end{aligned} \quad (38)$$

Combining (37) and (38), we have $x(t) = d p(t)$. Thus, we have $\mathcal{R}(x_0, t, \mathcal{T}) \subseteq d \mathcal{R}(p_0, t)$. Suppose (36) is not strongly accessible. Since $x \mapsto dx$ is a stretching transformation, we have $d \mathcal{R}(p_0, t)$ cannot contain an open subset of U . Hence, $\mathcal{R}(x_0, t, \mathcal{T})$ also cannot contain an open subset of U . It implies that the linear ICS (34) is not strongly accessible, which is a contradiction. Therefore, if (iii) holds, we have the corresponding impulse-free system (36) is strongly accessible. Then, by the theories of linear ODE, we have (36) is controllable and therefore (35) holds. It implies (ii) is true.

Hence, we obtain (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii). The proof is complete. $\blacksquare$

Remark 5. In fact, (35) is the same as the controllability rank condition in [14] for linear ICS (34). It is noted that, from [33], controllability is equivalent to strong accessibility for linear time-invariant control systems. We show that the equivalence remains valid for linear ICS (34) for the first time.

3.3. Observability

Next, we will investigate the observability of the nonlinear ICS (1). To this end, we present the following definitions inspired by [33].

Definition 8. For a point $x_0 \in U$ and a neighborhood V of x_0 , given a $T > 0$, a V -feasible control u is such a control that the trajectory remains in V , i.e., $x(t, x_0, u) \in V, \forall t \in [0, T]$.

Definition 9. For a given $\mathcal{T}$, if for any $t_f > 0$ and any V -feasible control u , the output $y(t, x_1, \mathcal{T}, u) = y(t, x_2, \mathcal{T}, u), t \in [0, t_f]$, then the states x_1, x_2 are said to be V -indistinguishable. The set of V -indistinguishable points from x_0 on the time interval $[0, t_f]$ with fixed impulse times is denoted by $ID_V(x_0, t_f, \mathcal{T})$.

Definition 10. The nonlinear ICS (1) is said to be *observable* at $x_0 \in U$ if there exists a neighborhood U' of x_0 such that $ID_{U'}(x_0, t_f, \mathcal{T}) = \{x_0\}$. If the system is observable at each x_0 then it is said to be observable.

Recall that for (strong) accessibility, the (strong) accessibility distribution of the nonlinear ICS (1) is essential. Similarly, for observability, we introduce the following observability distribution:

$$Q = \langle f, \varphi^\dagger, g_1, \dots, g_m | \text{span}\{dh_1, \dots, dh_p\} \rangle^\perp. \quad (39)$$

Lemma 4. Set

$$\begin{aligned} \Omega_0 &= \text{span}\{dh_1, \dots, dh_p\} \\ \Omega_i &= \Omega_{i-1} + L_f \Omega_{i-1} + L_{\varphi^\dagger} \Omega_{i-1} + \sum_{j=1}^m L_{g_j} \Omega_{i-1}, \quad i \geq 1. \end{aligned} \quad (40)$$

If there exists an integer i^* such that $\Omega_{i^*} = \Omega_{i^*+1}$, then $Q = \Omega_{i^*}^\perp$.

Lemma 4 can be obtained from Lemma 1.9.2 in [1]. Then the theorem on the observability is as follows.

Theorem 4. Suppose Q defined by (39) is nonsingular with dimension s . Then, for each $x_0 \in U$, it is possible to find a neighborhood U° of x_0 and a coordinates transformation $z = \Psi(x) = \text{col}(\psi_1(x), \dots, \psi_n(x))$ defined on U° . If

- (i) φ is a diffeomorphism from U° into its image such that Assumptions 2 and 3 are satisfied,
- (ii) there exists a U° -feasible control u ,

then we have following properties:

- (a) Any two initial states $x^a, x^b \in U^\circ$ with $\psi_i(x^a) = \psi_i(x^b), i = s+1, \dots, n$ produce identical output under any admissible input and impulsive effects.
- (b) There exists a neighborhood $Q \subseteq U^\circ$ of x_0 . For $0 < \tau_1 < \dots < \tau_l < t_f, 1 \leq l$, there is

$$ID_Q(x_0, t_f, \mathcal{T}) \subseteq S_{x_0} = \{x \in U^\circ | \psi_i(x) = \psi_i(x_0), i = s+1, \dots, n\}.$$

Proof. The proof of (a) is derived from the statement (b) of Theorem 1. Next, consider $t_1, \dots, t_k \geq 0$, such that $\tau_2 > \tilde{t}_k = t_1 + \dots + t_k \geq t_1 \geq \tau_1$ and define

$$\begin{cases} u_l(t) = u_l^1 & t \in [0, \tau_1], \\ u_l(t) = u_l^{1*} & t \in (\tau_1, t_1], \\ \dots & \\ u_l(t) = u_l^k & t \in (t_1 + \dots + t_{k-1}, t_1 + \dots + t_k], \end{cases} \quad (41)$$

$$\theta_l = f + \sum_{i=1}^m g_i u_i^l + \varphi^\dagger \delta(t - \tau_1), l = 1, 2, \dots, k,$$

$$\theta_1^* = f + \sum_{i=1}^m g_i u_i^{1*} + \varphi^\dagger \delta(t - \tau_1).$$

Then, the state reached at time $\tilde{t}_k$ starting from x_0 can be expressed as

$$x_0(\tilde{t}_k) = \Phi_{t_k}^{\theta_k} \circ \dots \circ \Phi_{t_2}^{\theta_2} \circ \Phi_{t_1-\tau_1}^{\theta_1^*} \circ \Phi_{\tau_1}^{\theta_1}(x_0).$$

If there is an initial state $x_1 \in ID_{U^\circ}(x_0, \tilde{t}_k, \mathcal{T})$, we have the output $h_j(x_0(\tilde{t}_k)) = h_j(x_1(\tilde{t}_k))$. Differentiating of both sides with respect to $t_k, t_{k-1}, \dots, t_1$ at respectively $t_k = t_{k-1} = \dots = t_2 = 0, t_1 = \tau_1$. It yields

$$L_{\theta_k} \dots L_{\theta_2} L_{\theta_1^*} h_j(\tilde{x}_0) = L_{\theta_k} \dots L_{\theta_2} L_{\theta_1^*} h_j(\tilde{x}_1), \quad (42)$$

where $\tilde{x}_0 = \Phi_{\tau_1}^{\theta_1}(x_0), \tilde{x}_1 = \Phi_{\tau_1}^{\theta_1}(x_1)$. Set $\gamma_1 = L_{\theta_{k-1}} \dots L_{\theta_2} L_{\theta_1^*} h_j$, we can obtain from (42) that $L_{\theta_k} \gamma_1(\tilde{x}_0) = L_{\theta_k} \gamma_1(\tilde{x}_1)$. Namely,

$$\begin{aligned} &L_f \gamma_1(\tilde{x}_0) + \sum_{i=1}^m L_{g_i} \gamma_1(\tilde{x}_0) u_i^k + L_{\varphi^\dagger} \gamma_1(\tilde{x}_0) \delta(0) \\ &= L_f \gamma_1(\tilde{x}_1) + \sum_{i=1}^m L_{g_i} \gamma_1(\tilde{x}_1) u_i^k + L_{\varphi^\dagger} \gamma_1(\tilde{x}_1) \delta(0). \end{aligned}$$

Thus, due to the arbitrariness of u_i^k , we have

$$L_{v_k} \gamma_1(\tilde{x}_0) = L_{v_k} \gamma_1(\tilde{x}_1), v_k \in \{f, g_1, \dots, g_m, \varphi^\dagger\}. \quad (43)$$

Set $\gamma_2 = L_{\theta_{k-2}} \dots L_{\theta_2} L_{\theta_1^*} h_j$. From (43), we similarly have

$$L_{v_k} L_{v_{k-1}} \gamma_2(\tilde{x}_0) = L_{v_k} L_{v_{k-1}} \gamma_2(\tilde{x}_1), v_k, v_{k-1} \in \{f, g_1, \dots, g_m, \varphi^\dagger\}.$$

This procedure can be iterated and finally we have

$$L_{v_k} \dots L_{v_1} h_j(\tilde{x}_0) = L_{v_k} \dots L_{v_1} h_j(\tilde{x}_1),$$

where $v_1, \dots, v_k$ are in the set $\{f, g_1, \dots, g_m, \varphi^\dagger\}$. From Lemma 1.9.4 of [1], there exists an open dense subset U^* of U° . Here are the two cases to consider based on whether x_0 belongs to U^* .

Case 1. For $x_0 \in U^*$, by Lemma 1.9.4 of [1], it is possible to find $\lambda_1, \dots, \lambda_{n-s}$ which have the form

$$\lambda_q = h_j \text{ or } \lambda_q = L_{v_r} \dots L_{v_2} L_{v_1} h_j, q = 1, \dots, n-s, \quad (44)$$

with $v_1, \dots, v_r \in \{f, g_1, \dots, g_m, \varphi^\dagger\}$ and $1 \leq j \leq p$, such that $Q^\perp = \text{span}\{d\lambda_1, \dots, d\lambda_{n-s}\}$. Since $\dim Q^\perp(x_0) = n-s$, there are $d\lambda_1(x_0), \dots, d\lambda_{n-s}(x_0)$ linearly independent. Define the mapping

$$\Lambda : (z_{s+1}, \dots, z_n) \mapsto (\lambda_1, \dots, \lambda_{n-s}),$$

where $\lambda_1, \dots, \lambda_{n-s}$ are only functions of $z_{s+1}, \dots, z_n$ from (11). Since $d\lambda_1(x_0), \dots, d\lambda_{n-s}(x_0)$ are linearly independent, we deduce that Λ has a nonsingular Jacobian matrix at $(\psi_{s+1}(x_0), \dots, \psi_n(x_0))$ and is locally injective. Then there exists a neighborhood $U' \subseteq U^\circ$ of x_0 with the property that Λ is injective for any $x \in U'$. Hence, for any other $x_1 \in U'$ such that $\lambda_q(\tilde{x}_0) = \lambda_q(\tilde{x}_1)$, there are $\psi_i(\tilde{x}_0) = \psi_i(\tilde{x}_1), i = s+1, \dots, n$. Then, we have $\psi_i(x_0) = \psi_i(x_1), i = s+1, \dots, n$. It implies that any initial state $x \in ID_{U'}(x_0, \tilde{t}_k, \mathcal{T})$ belongs to S_{x_0} . Namely, $ID_{U'}(x_0, \tilde{t}_k, \mathcal{T}) \subseteq S_{x_0}$.

Case 2. For $x_0 \notin U^*$. We first show that, for $t \in [0, T]$, and $T > \tau_1$, there is $\mathcal{R}^{U^\circ}(x_0, \mathcal{T}) \cap U^* \neq \emptyset$. If it is not true, from Theorem 2, we can find an r -dimensional submanifold $V \subseteq \mathcal{R}^{U^\circ}(x_0, \mathcal{T})$ and therefore $V \cap U^* = \emptyset$. For any choice of functions λ_q in (44), $d\lambda_1(x), \dots, d\lambda_{n-s}(x)$ are linearly dependent at $x \in V$. Thus, without loss of generality, we may assume that there exist $d < n-s$ functions $\eta_1, \dots, \eta_d$ still of the form (44), such that for some open subset V' of $V, x \in V'$, there are

$$\text{span}\{dh_1, \dots, dh_p\} \subseteq \text{span}\{d\eta_1, \dots, d\eta_d\},$$

$$dL_v \eta_q \in \text{span}\{d\eta_1, \dots, d\eta_d\}$$

for $q = 1, \dots, d, v \in \{f, g_1, \dots, g_m, \varphi^\dagger\}$, and $d\eta_1, \dots, d\eta_d$ are linearly independent. Hence, we can define a co-distribution as follows: $\Omega(x) = Q^\perp(x), x \notin V'$, and

$$\Omega(x) = \text{span}\{d\eta_1(x), \dots, d\eta_d(x)\}, x \in V'.$$

It is invariant under $f, g_1, \dots, g_m, \varphi^\dagger$ and contains $\text{span}\{dh_1, \dots, dh_p\}$. Note that, we have $\dim \Omega = d$ for $x \in V'$. On the other hand, by the definition, we have

$$Q^\perp = \langle f, \varphi^\dagger, g_1, \dots, g_m | \text{span}\{dh_1, \dots, dh_p\} \rangle$$

is the smallest co-distribution which contains $\text{span}\{dh_1, \dots, dh_p\}$ and is invariant under $f, \varphi^\dagger, g_1, \dots, g_m$. Since $Q^\perp$ is nonsingular with dimension $n-s > d$, we have Ω is smaller than $Q^\perp$, this is a contradiction. Hence, we have for any $x_0 \notin U^*$, there is $\mathcal{R}^{U^\circ}(x_0, \mathcal{T}) \cap U^* \neq \emptyset$ for $t \in [0, T]$ and $T > \tau_1$.

Next, let $x' = x(x_0, T_0, u)$ denote the state reached at time $t = T_0 \leq T$ from x_0 under the U° -feasible control u such that $x' \in U^*$. By case 1, we deduce that there exists a neighborhood $\hat{U}$ of x' such that $ID_{\hat{U}}(x', \tilde{t}_k, \mathcal{T}) \subseteq S_{x'}$. Due to φ being a diffeomorphism, we have the mapping

$$\Phi(x_0) := x(x_0, T_0, u) = \Phi_{T_0-T_1}^{\hat{\theta}} \circ \varphi \circ \Phi_{T_1}^{\hat{\theta}}(x_0)$$

is a local diffeomorphism, where $\hat{\theta}, \tilde{\theta}$ denote corresponding smooth vector fields. Thus, there exists a neighborhood $\bar{U}$ of x_0 whose (diffeomorphic) image under Φ is a neighborhood $\bar{U} \subseteq \hat{U}$ of x' . For any $\bar{x} \in ID_{\bar{U}}(x_0, \tilde{t}_k, \mathcal{T})$, we denote $x'' = x(\bar{x}, T_0, u)$. Then, there is $x'' \in ID_{\bar{U}}(x', \tilde{t}_k, \mathcal{T})$, since Φ is a local diffeomorphism. From the previous discussion we know that x'' and x' belong to $S_{x'}$. Due to Φ being a local diffeomorphism, it implies that $\bar{x}$ and x_0 belong to S_{x_0} . Namely, $ID_{\bar{U}}(x_0, \tilde{t}_k, \mathcal{T}) \subseteq S_{x_0}$.

Finally, set $Q = U'$ when $x_0 \in U^*$ or $Q = \bar{U}$ when $x_0 \notin U^*$, which is a neighborhood of x_0 . We have $ID_Q(x_0, \tilde{t}_k, \mathcal{T}) \subseteq S_{x_0}$ by the case 1 and 2. Note that $ID_Q(x_0, t_f, \mathcal{T}) \subseteq ID_Q(x_0, \tilde{t}_k, \mathcal{T})$ by the definition. Hence, the proof of (b) is complete. ■

Remark 6. In this theorem, we provide a geometric characterization of $ID_{U'}(x_0, t_f, \mathcal{T})$. If the nonlinear ICS (1) is impulse-free, then we can obtain Theorem 1.9.7 in [1] from this theorem.

Next, we have the following corollary from Theorem 4.

Corollary 4. Suppose Q defined by (39) is nonsingular. Then, for each $x_0 \in U$, it is possible to find a neighborhood U° of x_0 and a coordinates transformation $z = \Psi(x) = \text{col}(\psi_1(x), \dots, \psi_n(x))$ defined on U° . If conditions (i), (ii) of Theorem 4 are satisfied, and $\dim Q^\perp(x) = n$ for all $x \in U$, then we have the system (1) is observable.

Definition 11. The condition $\dim Q^\perp(x_0) = n$ is called the *observability rank condition* at x_0 . If $\dim Q^\perp(x_0) = n$ holds for any $x_0 \in U$, then the system (1) is said to satisfy the observability rank condition.

Remark 7. For linear ICS (34), it can be calculated from Lemma 4 that the observability rank condition is

$$\text{rank} \begin{bmatrix} C_c \\ C_c A_c \\ \vdots \\ C_c A_c^{n-1} \end{bmatrix} = n. \quad (45)$$

In fact, (45) is the same with observability rank condition in [14].

4. Examples

Example 1. In this example, we will verify the effectiveness of Theorem 1(a). Consider the system

$$\begin{cases} \dot{x}(t) = f(x(t)) + g(x(t))u(t), & t \neq \tau_k, \\ x(\tau_k^+) = \varphi(x(\tau_k)), & t = \tau_k, \end{cases} \quad (46)$$

with

$$f(x) = \text{col}(x_1 x_3 + x_2 e^{x_2}, x_3, x_4 - x_2 x_3, x_3^2 + x_2 x_4 - x_2^2 x_3),$$

$$g(x) = \text{col}(x_1, 1, 0, x_3),$$

$$\varphi(x) = \text{col}(x_1 + x_2 e^{x_2}, x_2, 2x_3, 2x_4).$$

By Lemma 3, we find $\Delta_0 = \text{span}\{g\}$, $\Delta_1 = \text{span}\{g, [f, g], [\varphi^\dagger, g]\}$, with

$$[f, g](x) = \frac{\partial g}{\partial x} f(x) - \frac{\partial f}{\partial x} g(x) = -\text{col}(e^{x_2}, 0, 0, 0),$$

$$[\varphi^\dagger, g](x) = \frac{\partial g}{\partial x} \varphi^\dagger(x) - \frac{\partial \varphi^\dagger}{\partial x} g(x) = -\text{col}(e^{x_2}, 0, 0, 0).$$

Proceeding further, setting $\theta = [f, g] = [\varphi^\dagger, g]$, we have

$$\Delta_2 = \Delta_1 + \text{span}\{[f, \theta], [g, \theta], [\varphi^\dagger, \theta]\}.$$

However, in this example we have $[f, \theta] = [g, \theta] = [\varphi^\dagger, \theta] = 0$. Therefore, the construction terminates, and $\langle f, \varphi^\dagger, g | \text{span}\{g\} \rangle = \Delta_1$, where the distribution Δ_1 has dimension 2 for all $x \in \mathbb{R}^4$. Notably, the system (46) without impulsive effects is equivalent to the system studied in Example 1.8.4 of [1]. Similar to the procedure in [1], we have the following transformation of coordinates $z = \Psi(x)$ in the state space:

$$z_1 = x_1, z_2 = x_2, z_3 = x_3, z_4 = x_4 - x_2 x_3.$$

Moreover, Assumption 1 is satisfied for such coordinates transformation, and Assumption 2 is satisfied by selecting $U_0 = \mathbb{R}^4$. Next we verify

that [Assumption 3](#) is satisfied. By calculation, we obtain

$$\Psi \circ \varphi(\Psi^{-1}(z(\tau_k))) = \begin{bmatrix} z_1(\tau_k) + z_2(\tau_k)e^{z_2(\tau_k)} \\ z_2(\tau_k) \\ 2z_3(\tau_k) \\ 2z_4(\tau_k) \end{bmatrix}, \quad (47)$$

and

$$\begin{aligned} z(\tau_k) + \frac{\partial \Psi}{\partial x} \varphi^\dagger(\Psi^{-1}(z(\tau_k))) &= z(\tau_k) + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -x_3(\tau_k) & -x_2(\tau_k) & 1 \end{bmatrix} \\ &\quad \times \begin{bmatrix} x_2(\tau_k)e^{x_2(\tau_k)} \\ 0 \\ x_3(\tau_k) \\ x_4(\tau_k) \end{bmatrix} \\ &= \begin{bmatrix} z_1(\tau_k) + z_2(\tau_k)e^{z_2(\tau_k)} \\ z_2(\tau_k) \\ 2z_3(\tau_k) \\ 2z_4(\tau_k) \end{bmatrix}. \end{aligned} \quad (48)$$

By (47) and (48), [Assumption 3](#) is satisfied.

In the new coordinates, the system becomes

$$\begin{cases} \dot{z}(t) = \begin{bmatrix} z_1 z_3 + z_2 e^{x_2} \\ z_3 \\ z_4 \\ 0 \end{bmatrix} + \begin{bmatrix} z_1 u \\ u \\ 0 \\ 0 \end{bmatrix}, & t \neq \tau_k, \\ z(\tau_k^+) = \begin{bmatrix} z_1(\tau_k) + z_2(\tau_k)e^{z_2(\tau_k)} \\ z_2(\tau_k) \\ 2z_3(\tau_k) \\ 2z_4(\tau_k) \end{bmatrix}, & t = \tau_k, \end{cases} \quad (49)$$

i.e., exactly in the form (8).

Example 2. In this example, we will illustrate the accessibility and observability of a nonlinear ICS on $\mathbb{R}^2$:

$$\begin{cases} \dot{x}_1(t) = u, & t \neq \tau_k, \\ \dot{x}_2(t) = 0, & t \neq \tau_k, \\ x_1(\tau_k^+) = x_1(\tau_k), & t = \tau_k, \\ x_2(\tau_k^+) = 1 + e^{x_1(\tau_k)}, & t = \tau_k, \\ y = x_2 \end{cases} \quad (50)$$

with $\tau_k = k, k \in \mathbb{Z}_+$.

Clearly, the accessibility distribution R is equal to the strong accessibility distribution P , which is spanned by $g = \text{col}(1, 0)$, $[\varphi^\dagger, g] = \text{col}(0, -2x_1)$. Therefore, $\dim R = \dim P = 2$ everywhere. By [Theorems 2](#) and [3](#), we have the system is (strongly) accessible. The reachable set at time $t_f > 1$ looks like in [Fig. 1](#), which contains an open subset of $\mathbb{R}^2$.

Then, we consider the observability. By [Lemma 4](#), we obtain

$$\Omega_0 = \text{span}\{dh_1\} = \text{span}\{(0, 1)\},$$

$$\Omega_1 = \Omega_0 + L_g \Omega_0 + L_{\varphi^\dagger} \Omega_0 = \text{span}\{(0, 1), (e^{x_1}, -1)\}.$$

Due to $\Omega_2 = \Omega_1$, the construction terminates. We have $\dim \Omega_1 = \dim \Omega^\perp = 2$. By [Theorem 4](#), this system is observable. Clearly, there exists a neighborhood U' of x_0 such that $ID_{U'}(x_0, t_f, \mathcal{T}) = \{x_0\}, t_f > 1$.

It should be noted that, without impulses, the system (50) is not (strongly) accessible and is not observable.

Example 3. We consider the impulsive multi-agent networks (MAN) in [11] without time delays. The dynamical process of agent i is

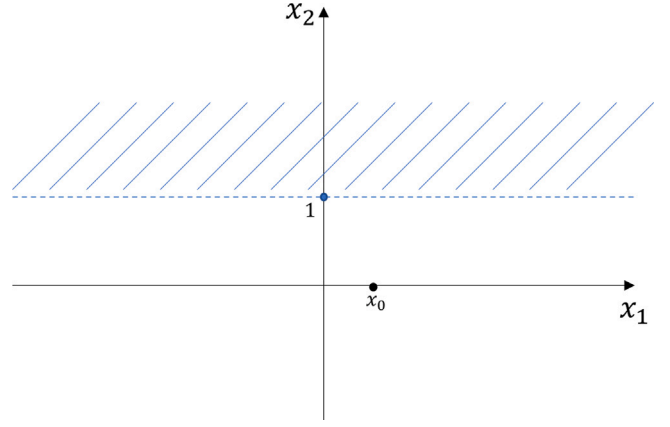


Fig. 1. Reachable set $\mathcal{R}(x_0, t_f, \mathcal{T})$ from x_0 at time t_f .

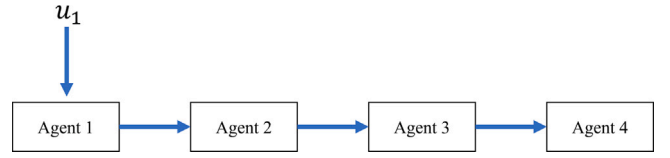


Fig. 2. The topology structure and external input node of MAN (52).

described by

$$\begin{cases} \dot{x}_i(t) = Ax_i(t) + \gamma_i Bu_i(t) + \sum_{j=1}^N \beta_{ij} F y_j, & t \neq \tau_k, \\ x_i(t^+) = d x_i(\tau_k), & t = \tau_k, \\ y_i(t) = E x_i(t), \\ x_i(t_0^+) = x_{i0}, \end{cases} \quad (51)$$

where $x_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}^p$ represent the state and output vectors of agent i ($i = 1, 2, \dots, N$); $d \in \mathbb{R}$ represents the impulsive strength; $u_i \in \mathbb{R}^d$ is the external input signal to agent i ; constant matrices A, B, E have suitable dimension; $F \in \mathbb{R}^{n \times p}$ represents the inner coupling matrix; and $\beta_{ij} \in \mathbb{R}$ represents the edge between agent i and agent j . Suppose that $\beta_{ii} = 0$ and $\beta_{ij} = 1$ if there exists an edge from agent j to agent i , otherwise $\beta_{ij} = 0$, for all $i, j = 1, 2, \dots, N$, and $\gamma_i = 1$ if agent i is under external input, otherwise $\gamma_i = 0$.

Let $X = [x_1^T, \dots, x_N^T]^T, Y = [y_1^T, \dots, y_N^T]^T, U = [u_1^T, \dots, u_N^T]^T$, and $X_0 = [x_{10}, x_{20}, \dots, x_{N0}]^T$ be the state, the output, the external control input, and the initial state of the whole network, respectively. Then, the compact form of MAN (51) is as follows:

$$\begin{cases} \dot{X} = (I_N \otimes A + \mathcal{A} \otimes FE)X + (\Gamma \otimes B)U, & t \neq \tau_k, \\ X(t^+) = dX(\tau_k), & t = \tau_k, \\ Y = (I_N \otimes E)X, \\ X(t_0^+) = X_0, \end{cases} \quad (52)$$

where $\otimes$ is Kronecker product. $\mathcal{A} = [\beta_{ij}] \in \mathbb{R}^{N \times N}$ and $\Gamma = \text{diag}(\gamma_1, \dots, \gamma_N)$. Similar to [11], we assume that there exist four agents, and the parameters of each agent in (51) are described as

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

and $d = 1.2, t_0 = 0, \tau_k = 0.5k, k \in \mathbb{Z}_+, t_f = 1.5$. The output matrix and inner coupling matrix are respectively assumed to be $E = [1, 0]$ and $F = [1, 1]^T$. The topology structure and external input node are presented in [Fig. 2](#).

By [Lemma 3](#), simple calculations lead to $\dim P = 8$ (equal to the dimension of X), which verifies the strong accessibility of MAN (52)

on $[0, 1.5]$ according to Corollary 3. Then by Proposition 1, we have MAN (52) is controllable on $[0, 1.5]$.

5. Conclusion

In this paper, some fundamental properties of nonlinear impulsive control systems (ICS) (1) including structural decompositions, accessibility and observability have been analyzed by differential geometric approach for the first time. We have proposed two structural decompositions of nonlinear ICS (1). By introducing the concepts of (strong) accessibility and observability distributions, the geometrical characterizations of the reachable and indistinguishable sets of nonlinear ICS (1) have been investigated, respectively. For nonlinear ICS (1), (strong) accessibility and observability rank conditions have been proposed, which relate to the (strong) accessibility and observability of such systems. When nonlinear ICS (1) degrades into a class of linear ICS (34), we have shown that the strong accessibility is equivalent to the controllability. Three examples have been provided to illustrate the effectiveness of the main theoretical results. This paper has provided a framework for the geometric analysis of nonlinear ICS (1), directing future research towards studying control problems in nonlinear ICS (1), such as stabilization, tracking, and regulation.

CRedit authorship contribution statement

Qinbo Huang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Lijuan Shen:** Validation, Visualization, Writing – review & editing. **Jitao Sun:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the editor and the reviewers for their constructive comments and suggestions which improved the quality of the paper.

Data availability

No data was used for the research described in the article.

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